MODULE 01 · FOUNDATIONS AND EXAM SKILLS - ALL DOMAINS

The Foundry mental model

One resource that governs, projects that build, and connections to everything Foundry does not own.

DOMAIN Foundations EXAM AI-103 MODULE 01 of 39

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The Foundry resource is where you govern. The project is where you build. Everything else is connected, and keeps its own rules.

Foundry as an airport

AT THE AIRPORT	IN FOUNDRY
The airport authority	The Foundry resource
Airport security and perimeter	Networking, keys, identity settings
Gates, shared by every terminal	Model deployments
Terminals	Projects
Airlines operating in a terminal	Agents
Rail and road links to the city	Connections to Storage, AI Search, Key Vault

ON THE EXAM

The rail operator runs its own ticket barriers. Securing the airport does not secure the train.

Resource or project?

THE FOUNDRY RESOURCE

Governance boundary

- Networking and private access
- Security and encryption settings
- Model deployments
- Connections to other Azure services

A FOUNDRY PROJECT

Development boundary

- Agents
- Files and uploaded data
- Evaluations and their results
- Reuses the resource's deployments

THE TIEBREAKER

Configured once for everyone: resource. Belongs to one team's work: project.

Not a new kind of Azure resource

- A Foundry resource is **Microsoft.CognitiveServices/accounts**, of kind **AIServices**.
- A project is a subresource: **accounts/projects**.
- Same provider as Azure OpenAI, Speech, Language and Vision.
- So management APIs, RBAC actions, networking and Azure Policy aliases line up across them.

> ON THE EXAM
>
> Upgrading an Azure OpenAI resource to Foundry keeps its existing custom policies and RBAC actions.

Connected resources keep their own rules

- Storage, Key Vault and AI Search are **separate Azure resources** with their own governance boundaries.
- Private networking on Foundry does **not** make a connected storage account private.
- Plan networking, access and compliance for **each** connected resource separately.
- By default, key-based connection secrets are held in a managed Key Vault. You can bring your own.

> ON THE EXAM
>
> If a question says the data must stay private, check the storage and search services, not just Foundry.

A deployment is a named way to reach a model

- Code calls the **deployment name**, not the model's catalogue name.
- Deploy the same model **more than once** to test different settings, such as content filters.
- Content filters are configured **per deployment**.
- Deployment types choose where data is processed: global, data zone or single region.

> ON THE EXAM
>
> Module 04 covers deployment types and billing. For now: a deployment is a gate, and you can build several gates to the same runway.

Which endpoint, with which SDK

ENDPOINT	USE IT FOR
`...services.ai.azure.com/api/projects/NAME`	Foundry SDK: agents, evaluations, models, platform tools
`...openai.azure.com/openai/v1`	OpenAI SDK: full OpenAI surface, embeddings, lowest latency
`...services.ai.azure.com/anthropic`	Anthropic SDK, for Claude models deployed in Foundry
`...cognitiveservices.azure.com`	Foundry Tools SDKs: Vision, Language, Content Safety and more

ON THE EXAM

The project endpoint is the default for new work. A standalone Azure OpenAI resource only offers the openai/v1 endpoint.

Two clients, two jobs

THE PROJECT CLIENT

Foundry-native setup

- List connections
- Read project properties
- Enable tracing

THE OPENAI-COMPATIBLE CLIENT

Running things

- Responses API calls to models
- Agents
- Evaluations and fine-tuning

WATCH FOR THIS

Embeddings are the exception. The project endpoint does not currently route them, so call openai/v1.

Foundry Tools: the prebuilt specialists

FOUNDRY TOOL	TYPICAL JOB
Azure Vision	Analyse images; read text from images
Azure Speech	Speech to text, text to speech, speech translation
Azure Language	PII detection, entities, language detection
Azure Translator	Text and document translation
Azure Content Safety	Detect harmful text and images
Content Understanding	Extract structure from documents, images, audio, video

ON THE EXAM

On AI-103 these are tools an app or agent uses, not the centre of the design. Translator is reached at resource level, not project level.

Azure Language: what stays, what goes

CAPABILITY	STATUS
Language detection, PII detection	Continued support
Prebuilt NER, custom NER, Text Analytics for Health	Continued support
Sentiment and opinion mining, key phrases	Retires 31 March 2029
Summarisation, custom text classification	Retires 31 March 2029
CLU, custom question answering, orchestration	Retires 31 March 2029
Entity linking	Retires 31 March 2029

ON THE EXAM

Not everything classic is leaving. Deterministic PII detection is still a first-class Foundry Tool answer.

Three ways to run an agent

APPROACH	WHAT YOU PROVIDE	CHOOSE IT WHEN
Prompt agent	Instructions, a model, tools	You want the least to manage
Hosted agent	Your own code as a container	You need custom logic or a framework
Your own runtime	Code calling the project's Responses API	The agent already lives in your app

ON THE EXAM

Choose the least complex option that meets the requirement. Modules 17 to 20 build all three.

Until they are published, every agent in a project shares **one identity**. Different access needs mean different projects.

Which setup is the exam's default?

CLASSIC

Still exists, not the direction

- Hub-based projects, built on Azure Machine Learning workspaces
- Foundry classic portal only
- Standalone Azure OpenAI: openai/v1 endpoint only

CURRENT

Foundry resource and projects

- Supported in the new Foundry portal
- Agents, evaluations, models, tools in one place
- Azure OpenAI resources can upgrade in place

HONEST EXCEPTION

A standalone Azure OpenAI resource can still suit completions-only work with no agents or evaluations.

ALL DOMAINS / JUST RENAMED

The Foundry roles, recently renamed

ROLE	WHAT IT ALLOWS
Foundry Agent Consumer	Call agent endpoints. Nothing else
Foundry User	Build and test in a project. Formerly Azure AI User
Foundry Project Manager	Manage projects, publish agents, assign Foundry User
Foundry Account Owner	Create projects, manage models. Cannot build
Foundry Owner	Everything, including building. Highly privileged

ON THE EXAM

Every Azure AI role became a Foundry role with the same ID and permissions. Questions written before the rename use the old names.

Things Foundry does not do for you

- **No automatic cross-region failover.** Multi-region means separate resources and routing in your app.
- Models, deployment types and features **vary by region**. Check before provisioning.
- Some network-isolated setups need the **SDK or CLI**, not the portal.
- Agent state uses Microsoft-managed storage by default. **Standard setup** brings your own.

> ON THE EXAM
>
> When a scenario demands resilience or data residency, the answer is design you add, not a switch you flip.

Follow one request through Foundry

01
Authenticate
A Microsoft Entra token for an identity with the right role.

02
Endpoint
The project endpoint, or openai/v1.

03
Deployment
Named in the request; resolves to a model.

04
Guardrails in
Input is checked before the model sees it.

05
Model
Generates, possibly calling tools.

06
Guardrails out
Output is checked, then traced and returned.

ON THE EXAM

Nearly every exam question sits at one of these six steps. Find the step, and half the options fall away.

> A retailer's platform team has bought model capacity for every team to share. Marketing is building a copywriting agent; HR is building a policy assistant.
>
> Both agents will be in development for months before publishing. The HR agent must read a confidential storage account; the marketing agent must never reach it. Security wants networking managed in one place.

How should the platform team structure Foundry?

A One resource and one project, with the storage role granted only to the HR agent's identity.

B Two Foundry resources, one per team, each with its own model deployments and networking.

C One resource and two projects, granting storage access only to the HR project's identity.

D A hub-based project per team, sharing a standalone Azure OpenAI resource.

◆ PAUSE AND DECIDE

How should the platform team structure Foundry?

A One project, storage role granted to the HR agent's identity.

Unpublished agents in a project share one identity, so marketing's agent gets the access too.

B Two resources, one per team.

It isolates access, but splits the shared capacity and puts networking in two places.

C One resource, two projects, access granted to the HR project's agent identity only.

Shared deployments and central networking on the resource; isolation at the project identity.

D Hub-based projects and standalone Azure OpenAI.

The classic setup. It is not supported in the new portal and is not where Microsoft is investing.

THE SIGNAL

'In development for months' is the planted detail. Unpublished agents share their project's identity, so isolation must be a project.

> A developer connects an app to a Foundry project using AIProjectClient and the project endpoint, authenticating with DefaultAzureCredential. Calls to a chat model through the Responses API succeed.
>
> They add a call to generate embeddings for a search index, using the same project endpoint. That call fails, while chat calls keep working.

What should the developer do?

A Send embedding requests to the resource's openai/v1 endpoint instead.

B Assign the developer the Foundry Account Owner role on the Foundry resource.

C Deploy the embedding model inside the project instead of on the Foundry resource.

D Replace DefaultAzureCredential with an API key for the project endpoint.

◆ PAUSE AND DECIDE

What should the developer do?

A Send embeddings to openai/v1 with the OpenAI SDK.

The project endpoint does not currently route embedding requests. Microsoft directs them to openai/v1.

B Assign Foundry Account Owner.

Chat calls already succeed, so permissions are not the problem. This role is also far too broad.

C Deploy the model inside the project.

Deployments live on the resource and projects reuse them. There is no project-level deployment to make.

D Switch to an API key.

Authentication already works for chat. A key cannot fix routing, and keys bypass role-based access entirely.

THE SIGNAL

'Chat keeps working' rules out identity, role and credentials in one sentence. What is left is routing.

> An insurer has run a standalone Azure OpenAI resource for two years. Its cloud team has attached custom Azure Policy definitions and custom RBAC role assignments to it, and audit depends on them.
>
> The business now wants agents and model evaluations. The cloud team fears that moving to Foundry means rebuilding the governance audit relies on.

What should the insurer do?

A. Create a new Foundry resource, recreate every policy and role assignment on it, then migrate workloads.

B. Keep the Azure OpenAI resource and add a hub-based project that connects to it.

C. Stay on the standalone resource and build agents through its openai/v1 endpoint.

D. Upgrade the Azure OpenAI resource to Foundry, keeping its existing custom policies and RBAC actions.

◆ PAUSE AND DECIDE

What should the insurer do?

A New resource, recreate governance, migrate.

It would work, but it is exactly the rebuild the team wants to avoid, with audit risk during the gap.

B Add a hub-based project.

The classic setup, unsupported in the new portal. It adds a second governance surface.

C Build agents through openai/v1.

A standalone Azure OpenAI resource does not provide agents or evaluations, the two things needed.

D Upgrade in place to Foundry.

Same resource provider, so existing custom policies and RBAC actions continue to apply.

THE SIGNAL

The fear in the question is the clue. Foundry shares the Cognitive Services provider, which is why governance carries over.

> A contractor joins a team for three months to build and evaluate agents in an existing Foundry project. The project and its model deployments are already set up.
>
> Security policy requires least privilege. The contractor must not be able to create projects, add connections, or change deployments.

Which role assignment should the team give the contractor?

A Foundry Account Owner on the Foundry resource.

B Foundry User on the project.

C Contributor on the resource group that contains the Foundry resource.

D Foundry Project Manager on the project.

◆ PAUSE AND DECIDE

Which role assignment should the team give the contractor?

A Foundry Account Owner on the resource.

Creates projects and manages models, both forbidden. It also has no data actions, so it cannot even build.

B Foundry User on the project.

Microsoft's role for day-to-day development, at the narrowest scope that does the job.

C Contributor on the resource group.

Broad and at a wide scope. It can create projects and deploy models, yet has no data actions to build agents.

D Foundry Project Manager on the project.

The right scope, but it can publish agents and grant roles to others, which is more than needed.

THE TIEBREAKER

Least privilege has two dials: the role and the scope. Turn both down until the job only just fits.

The mental model in five lines

- **Resource governs:** networking, security, deployments, connections.
- **Project builds:** agents, files, evaluations. One shared agent identity until publishing.
- **Connected services keep their own rules.** Secure each one.
- **Project endpoint by default;** openai/v1 for embeddings and pure OpenAI clients.
- **Least complex agent that works:** prompt, then hosted, then your own runtime.

> ON THE EXAM
>
> Next, module 02 opens Domain 1 with the first real decision on any AI project: which model.